BRIGHT PRIVATE CENTRE

Chapter:3

3.1

1. $(a+b)^2 = a^2 + 2ab + b^2$

2. $(a-b)^2 = a^2 - 2ab + b^2$

3. $a^2 + b^2 = (a+b)^2 - 2ab$

4. $a^2 + b^2 = (a-b)^2 + 2ab$

5. $a^2 - b^2 = (a+b)(a-b)$

6. $4ab = (a+b)^2 - (a-b)^2$

7. $ab = (\frac{a+b}{2})^2 - (\frac{a-b}{2})^2$

8. $(a-b)^2 = (a+b)^2 - 4ab$

9. $(a+b+c)^2 = a^2 + b^2 + c^2 + 2ab + 2bc + 2ca$

10. $2(ab+bc+ca) = (a+b+c)^2 - (a^2+b^2+c^2)$

3.2

1. $(a+b)^3 = a^3 + 3a^2b + 3ab^2 + b^3 = a^3 + b^3 + 3ab(a+b)$

2 . $(a-b)^3 = a^3 - 3a^2b + 3ab^2 - b^3 = a^3 - b^3 - 3ab(a-b)$

3. $a^3 + b^3 = (a+b)^3 - 3ab(a+b)$

4. $a^3 - b^3 = (a-b)^3 + 3ab(a-b)$

5. $a^3 + b^3 = (a+b)(a^2 - ab + b^2)$

6. $a^3 - b^3 = (a-b)(a^2 + ab + b^2)$

Md. Raihan Ahamed

Dept. of Mathamatics,

Jagannath university,Dhaka

Mob. 01640096834